learn the appropriate edges between the nodes. Such a representation would give us much more insight into the mechanics of our machine learning model than would a conventional convolutional neural network or the like.

Different neural network architectures such as convolutional, recurrent, or attention have different *inductive biases* that can improve learning when those biases are accurate. *Inductive reasoning* is when you observe some data and infer a more general pattern or rule from it. *Deductive reasoning* is what we do in mathematics when we start with some premises and by following logical rules assumed to be true, we can make a conclusion with certainty.

For example, the syllogism “All planets are round. Earth is a planet. Therefore, the Earth is round” is a form of deductive reasoning. There is no uncertainty about the conclusion if we assume the premises to be true.

Inductive reasoning, on the other hand, can only lead to probabilistic conclusions. Inductive reasoning is what you do when you play a game like chess. You cannot deduce what the other player is going to do; you have to rely on the available evidence and make an inference. Biases are essentially your expectations before you have seen any data. If you always expected your chess opponent, no matter who it was, to make a particular opening move, that would be a strong (inductive) bias.

Biases are often talked about in the pejorative sense, but in machine learning architectural biases are essential. It is the inductive bias of compositionality, i.e., that complex data can be decomposed into simpler and simpler components in a hierarchical fashion, that makes deep learning so powerful in the first place. If we know the data is images in a grid-like structure, we can make our models biased toward learning local features as convolutional neural networks do. If we know our data is relational, a neural network with a relational inductive bias will improve performance.

10.1.1. Invariance and equivariance